

The equiangular-frame lattice question was completely characterized

Literature-status packet for arXiv:1607.05350

17 August 2026

Status. Literature already answered: full characterization of the lattice-existence part of the source question.

Source question

Albrecht Böttcher, Lenny Fukshansky, Stephan Ramon Garcia, Hiren Maharaj, and Deanna Needell ask in the introduction of *Lattices from tight equiangular frames* (arXiv:1607.05350; locally compiled source PDF page 1):

When is the integer span $\Lambda(G)$ of a real unit ETF a lattice?

Writing its Gram matrix as

$$G^T G = I + \alpha^{-1} C$$

with C a sign matrix, source Proposition 2.1 proves the necessary condition that α be rational. The paper then treats particular families and low-dimensional cases, but does not state the converse.

Exact later answer

Böttcher and Fukshansky explicitly correct this omission in *Lattices and quadratic forms from tight frames in Euclidean spaces* (arXiv:1704.07735; published as an addendum in *Linear Algebra and its Applications* 531 (2017), 592–601). Its abstract says the earlier paper overlooked a simple argument in the reverse direction. Proposition 1 proves that any rational spanning family has lattice integer span, since after clearing a common denominator d , the norm form satisfies

$$d\|Ga\|^2 \in \mathbb{Z}_{\geq 0} \quad (a \in \mathbb{Z}^n).$$

For a unit ETF, rationality of the Gram matrix is equivalent to rationality of α (and the ETF parameter restrictions force such an α to be a positive integer). Theorem 3 therefore gives

$\Lambda(G)$ is a lattice $\iff \alpha \in \mathbb{Z}_{>0} \iff Q_G$ has separated values.

Scope and provenance

This completely answers the source’s lattice-existence question for real unit equiangular tight frames. It does not settle the source’s separate, open-ended request to determine all geometric properties of every resulting lattice. The identification is explicit: the addendum names the source paper, describes the missing reverse implication, and packages the equivalence as Theorem 3.

Search and file evidence

The run indexes were checked for both titles, arXiv identifiers, integer span, rational, and equiangular tight frame. Exact-title and keyword web searches found arXiv:1704.07735, whose full-text Proposition 1 and Theorem 3 were inspected through the arXiv/author-hosted primary text. The source paper was compiled locally. A local supporting-PDF download was attempted but unavailable because this runtime’s external-download approval quota was exhausted; the stable arXiv record is <https://arxiv.org/abs/1704.07735>.

References

1. A. Böttcher, L. Fukshansky, S. R. Garcia, H. Maharaj, and D. Needell, *Lattices from tight equiangular frames*, Linear Algebra Appl. 510 (2016), 395–420; arXiv:1607.05350.
2. A. Böttcher and L. Fukshansky, *Lattices and quadratic forms from tight frames in Euclidean spaces*, Linear Algebra Appl. 531 (2017), 592–601; arXiv:1704.07735.